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25 - 04 - 2026

SMEZY

SME FINANCIAL DECISION INTELLIGENCE SYSTEM

BY CTRL+SHE

THE TEAM BEHIND

SMEZY

FRONTEND & MEDIA LEAD

AINA WAFIYAH

BACKEND & DOCUMENTATION

AINA ALESHA

BACKEND & SYSTEM ARCHITECTURE

DHANIA NAFEESA

DOCUMENTATION & ANALYSIS

AINA JAMAL

FULL-STACK SUPPORT

AQASYAH FATIN AFRINA

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PROBLEMS

Lack of Cash Flow Visibility

SME owners do not have a clear and real-time view of their cash flow, making it difficult to assess their financial position accurately.

Fragmented Financial Data

Financial information is scattered across multiple sources, leading to inefficiencies in tracking and analyzing overall business performance.

Poor Decision-Making Without Simulation Tools

The absence of tools to simulate financial outcomes causes SME owners to rely on assumptions, resulting in ineffective or risky financial decisions.

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SOLUTIONS

This project proposes an AI-powered decision intelligence system that supports SME owners in managing their finances more effectively.

Target User

- SME Owners
- Entrepreneurs
- Small Business Manager
- Financial Staff in SMEs
- Business Consultants

AI-Powered Financial Simulation and Decision Support

The system enables SME owners to simulate financial scenarios (e.g., delaying payments, collecting invoices, taking loans) and provides data-driven outcomes with clear recommendations.

Integrated and Comprehensive Data Analysis

The system consolidates structured and unstructured financial data into a unified platform, offering a complete and accurate view of the business's financial health.

Insight Generation and Predictive Capabilities

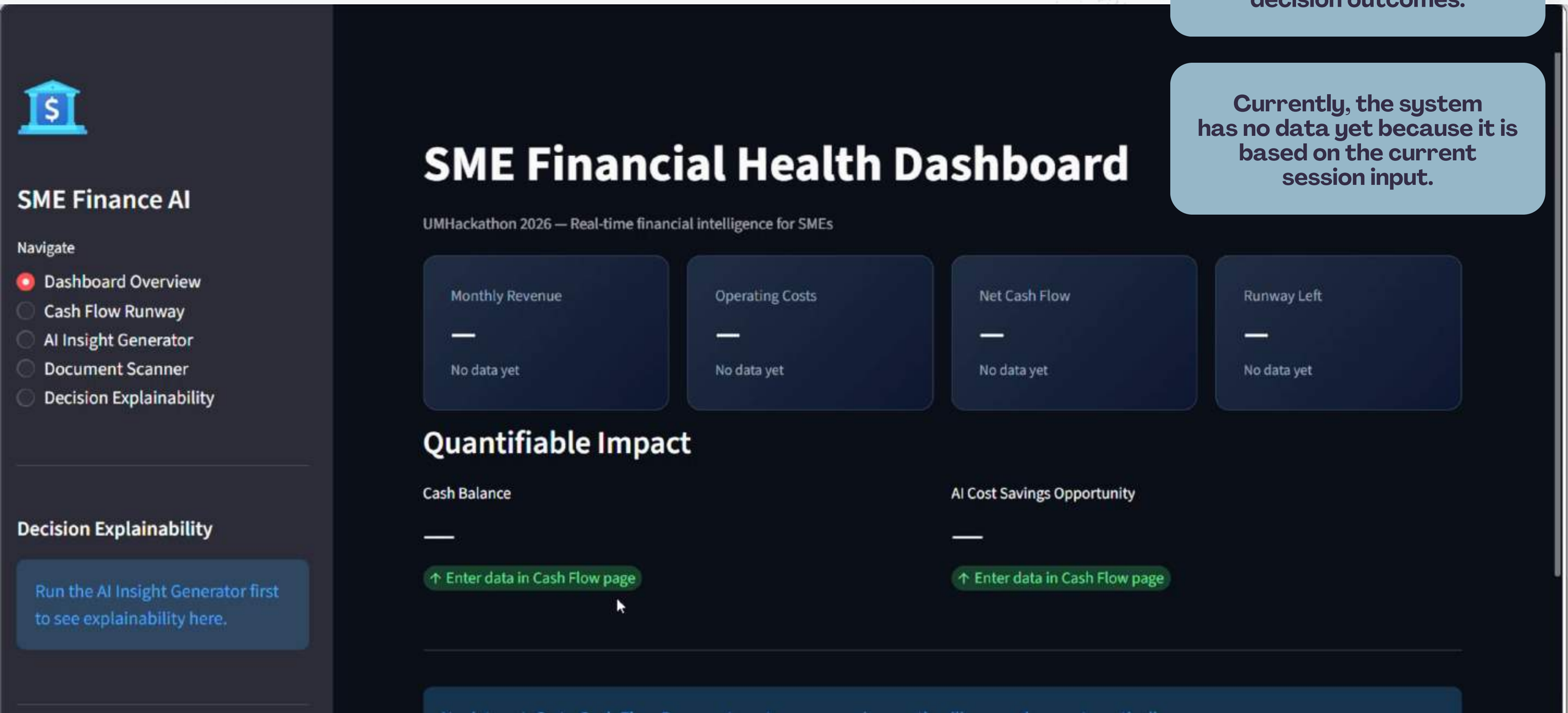
The system delivers actionable insights such as cash flow predictions, spending analysis, and document intelligence, helping users make informed and strategic financial decisions.

SMEZY FEATURES OVERVIEW

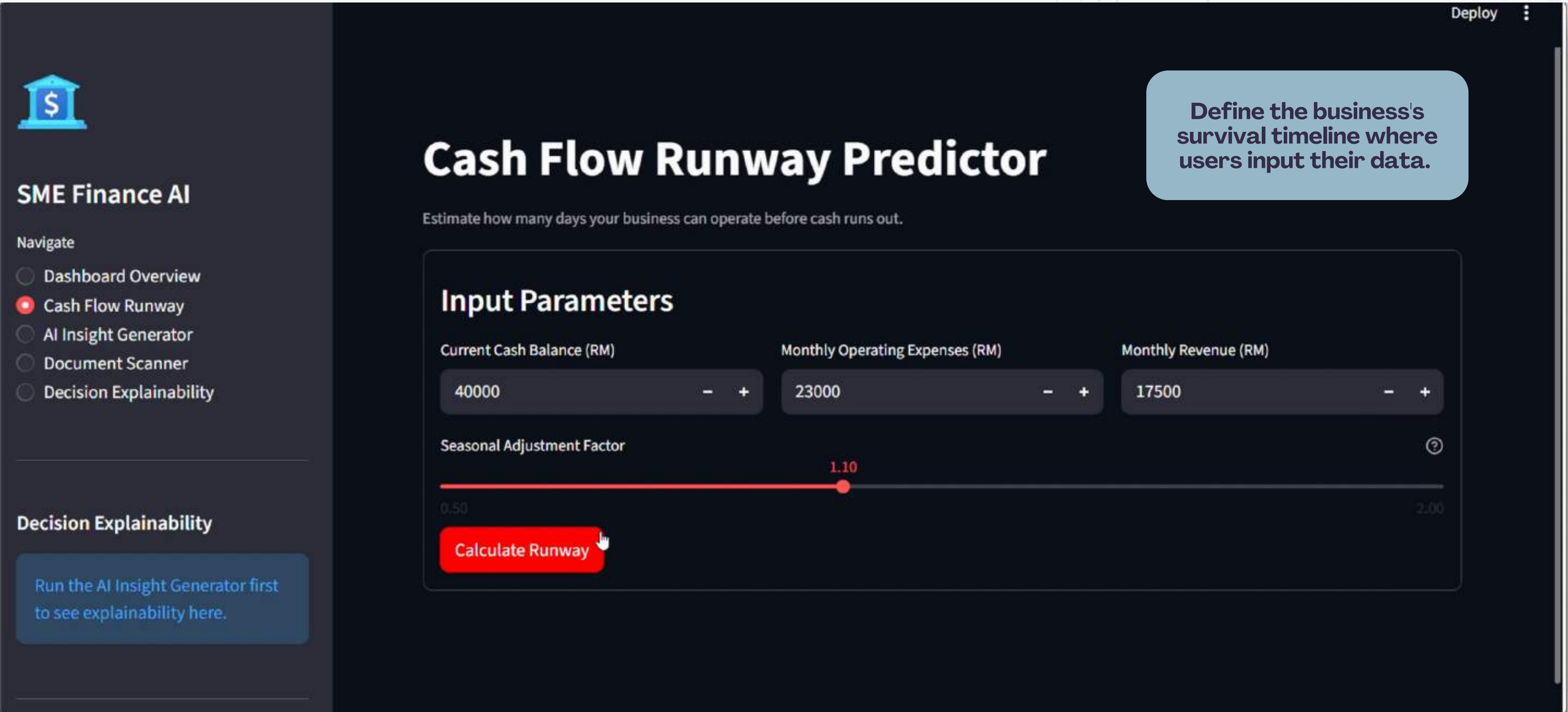
I. DASHBOARD OVERVIEW

Acts as the main overview where users can see their financial summary and decision outcomes.

Currently, the system has no data yet because it is based on the current session input.

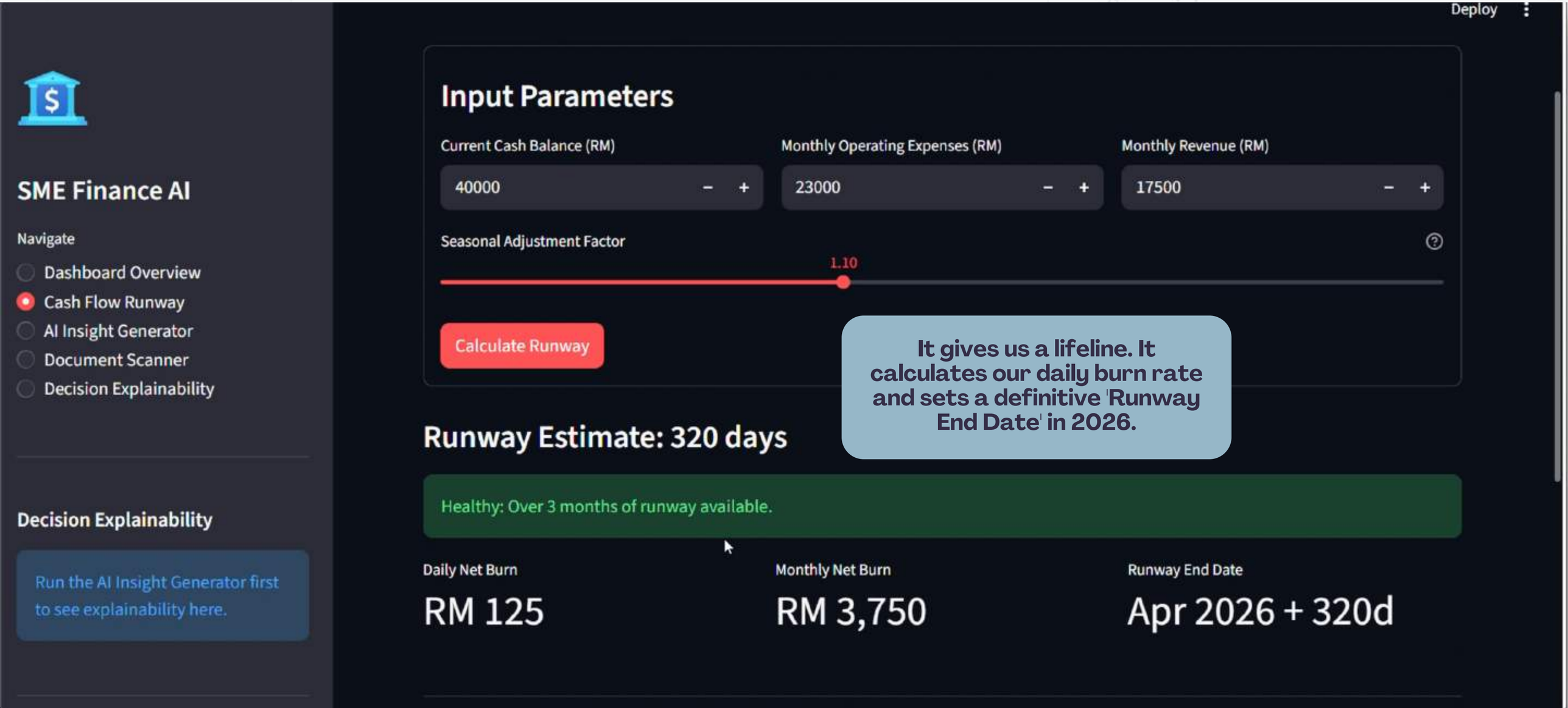


2. CASH FLOW RUNWAY



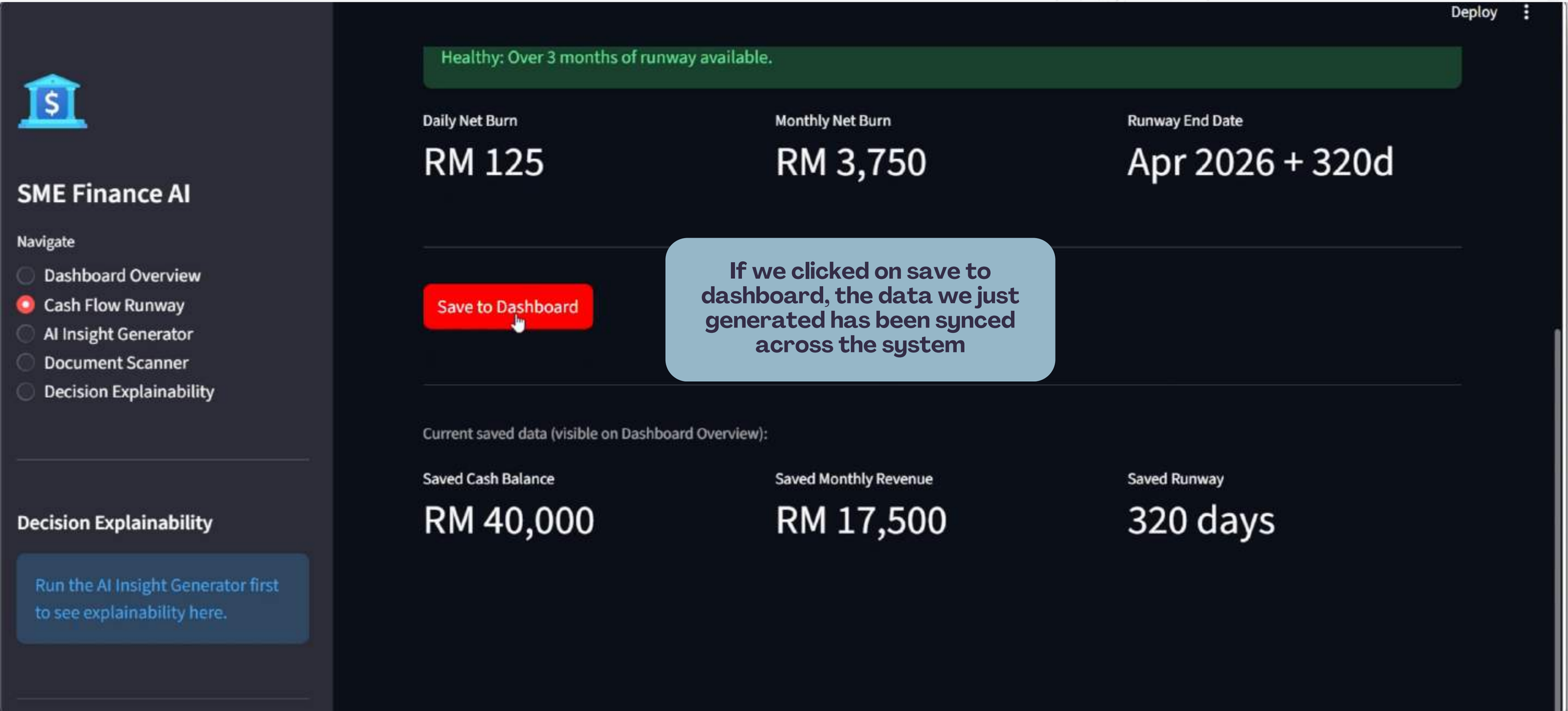
Define the business's survival timeline where users input their data.

2. CASH FLOW RUNWAY



It gives us a lifeline. It calculates our daily burn rate and sets a definitive 'Runway End Date' in 2026.

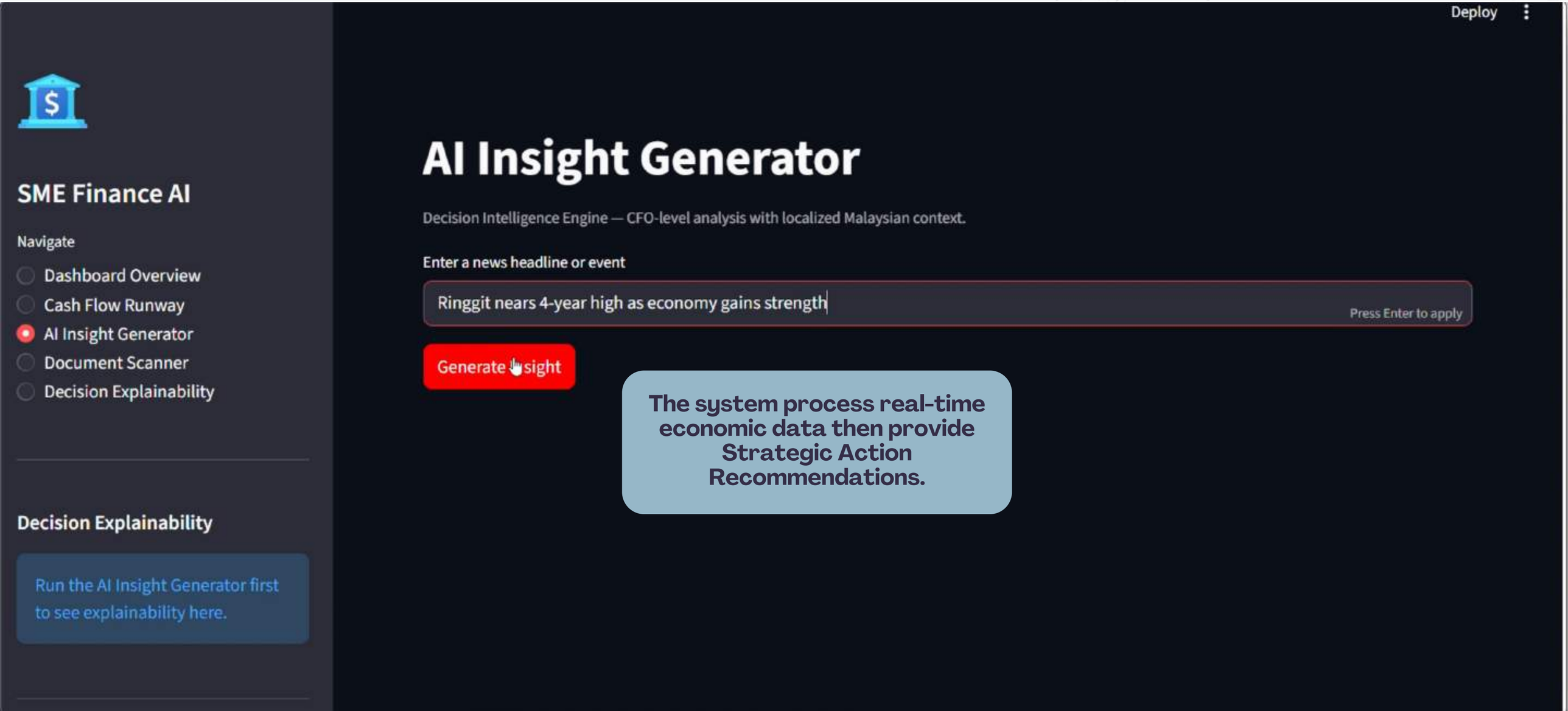
2. CASH FLOW RUNWAY



I. DASHBOARD OVERVIEW

See, now our dashboard have summary!

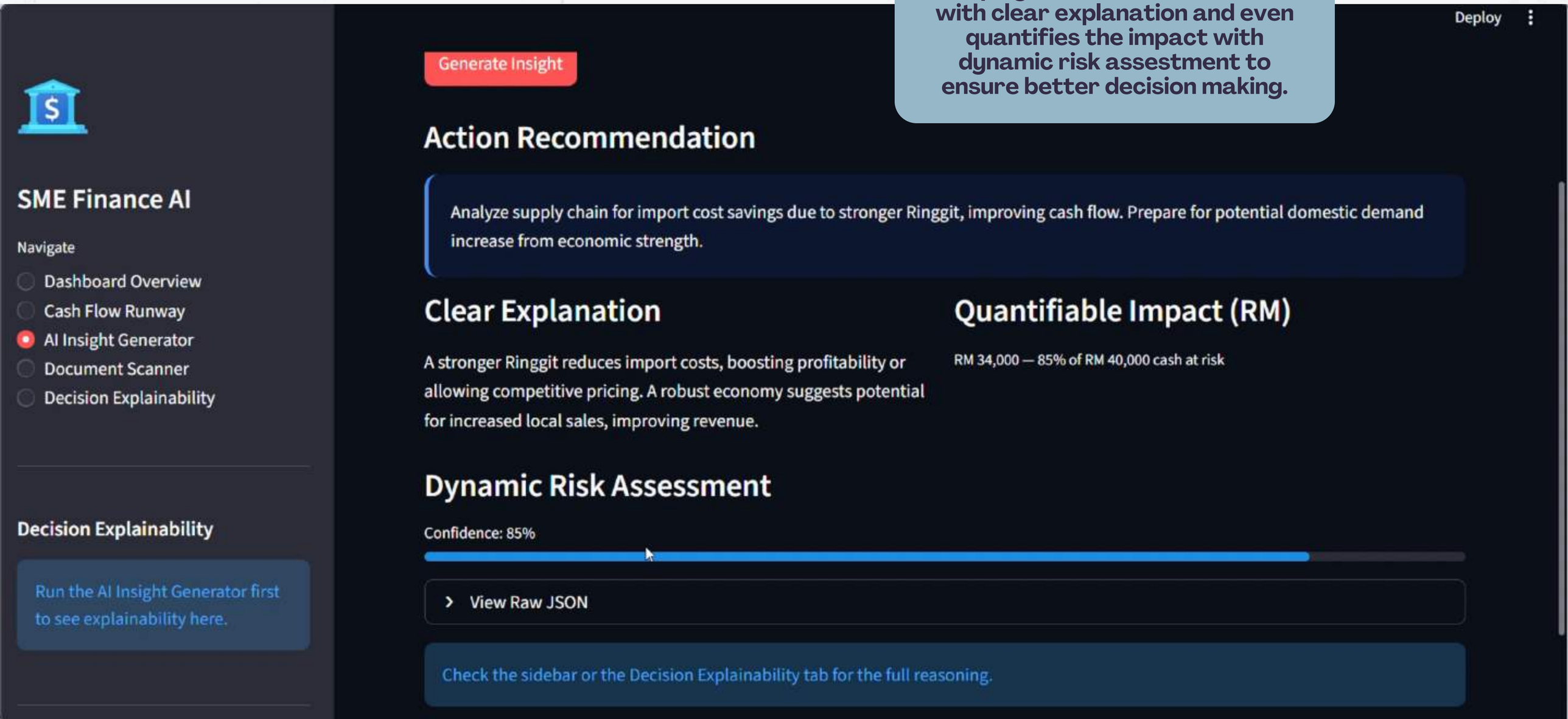
3. AI INSIGHT GENERATOR



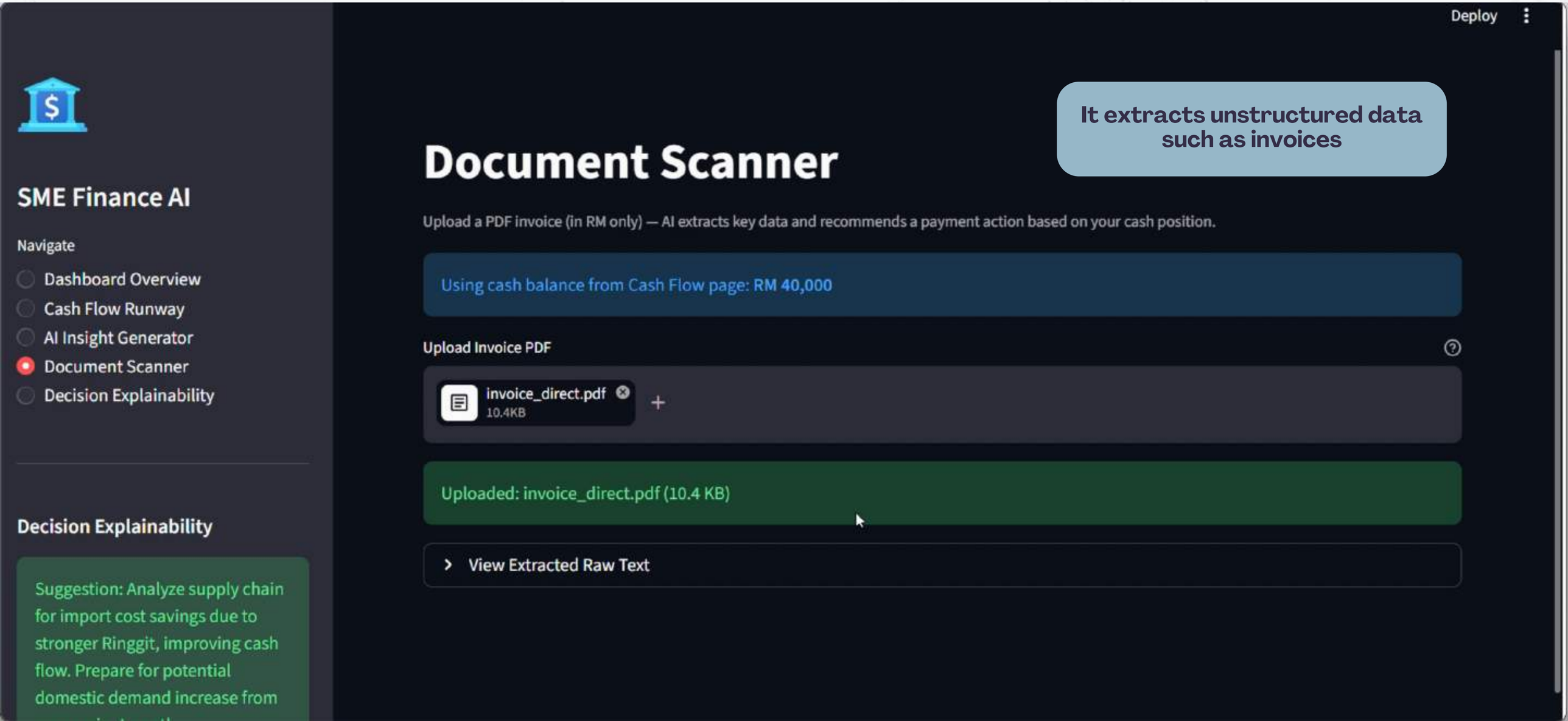
The system process real-time economic data then provide Strategic Action Recommendations.

3. AI INSIGHT GENERATOR

It display action recommendation with clear explanation and even quantifies the impact with dynamic risk assestment to ensure better decision making.



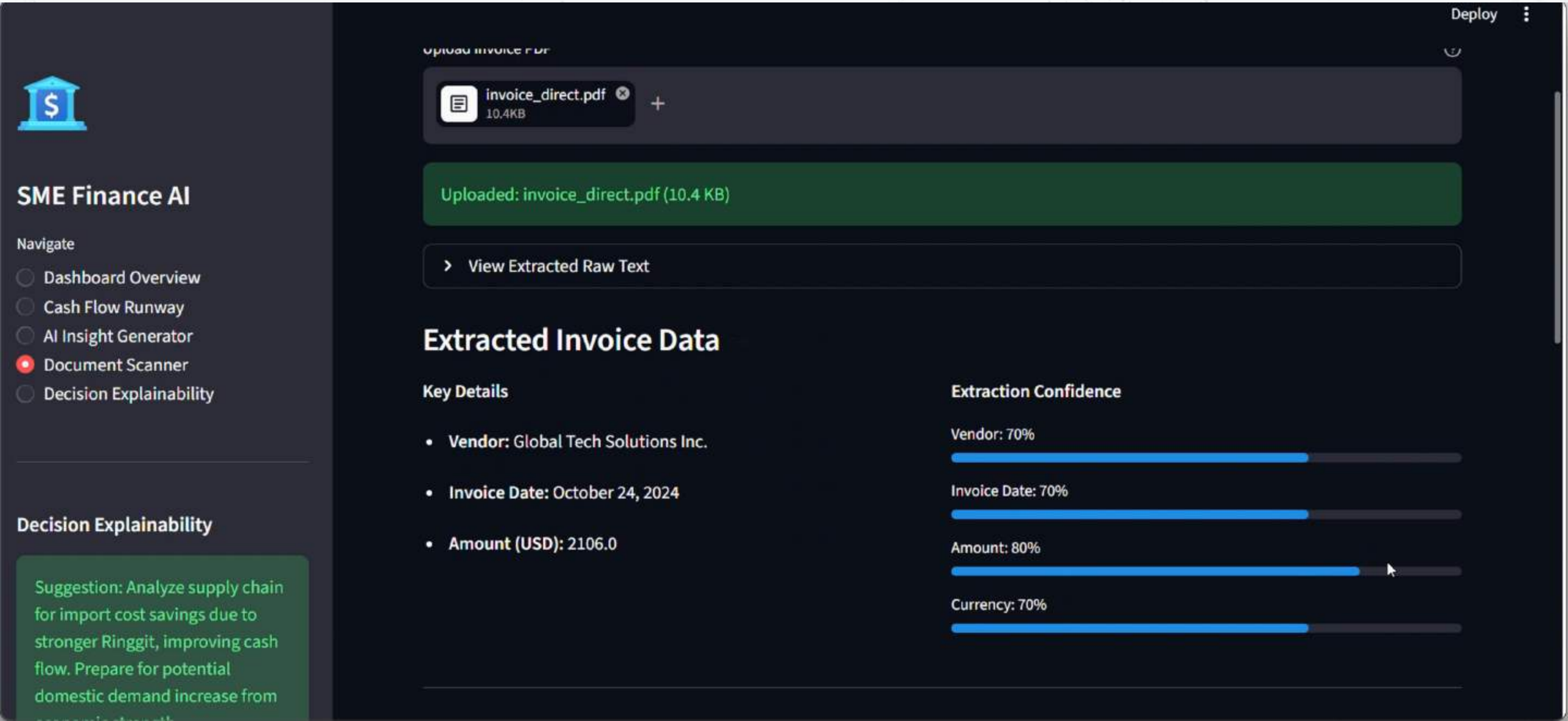
4. DOCUMENT SCANNER



It extracts unstructured data such as invoices

4. DOCUMENT SCANNER

The system lists the key details which are vendor, invoice date and amount



4. DOCUMENT SCANNER

The system also display strategic insight where it retrieves the current cash position and performs a Liquidity Ratio Analysis and issues an autonomous recommendation based on it.

Deploy

- Invoice Date: October 24, 2024
- **Amount (USD):** 2106.0

Invoice Date: 70%

Amount: 80%

Currency: 70%

SME Finance AI

Navigate

- Dashboard Overview
- Cash Flow Runway
- AI Insight Generator
- Document Scanner
- Decision Explainability

Strategic Insight

USD 2,106.00 invoice is 5% of USD 40,000 cash — manageable.

Extraction method: Gemini | Text length: 402 chars

Payment Recommendation

Pay Now

Risk Level: LOW

Decision Explainability

Suggestion: Analyze supply chain for import cost savings due to stronger Ringgit, improving cash flow. Prepare for potential domestic demand increase from

5. DECISION EXPLAINABILITY PANEL

Deploy

SME Finance AI

Navigate

- Dashboard Overview
- Cash Flow Runway
- AI Insight Generator
- Document Scanner
- Decision Explainability

Decision Explainability

Suggestion: Pay Now — Invoice of RM 2,106.00

> Why did the AI suggest this?

Decision Explainability Panel

Understand the reasoning behind every AI-generated financial recommendation.

It explains the full reasoning of AI by using the Reasoning Chain.

Current AI Suggestion

Pay Now — Invoice of RM 2,106.00

Source: Document Scanner

Reasoning Chain

Step 1: Document Parsing — Extracted unstructured text from the uploaded PDF invoice.

Step 2: AI Field Extraction — Identified Vendor: Global Tech Solutions Inc., Amount: RM 2,106.00, Due: None using AI (Gemini → Z.AI → local).

Step 3: Cash Position Lookup — Retrieved current Cash Balance of RM 40,000 from the Cash Flow Runway page.

Step 4: Ratio Analysis — Calculated that an RM 2,106.00 outflow represents 5% of your current liquidity (RM 40,000).

5. DECISION EXPLAINABILITY PANEL

The panel documents every logical step from Parsing the PDF, to Decision Logic by providing this level of transparency,

Deploy

local).

Step 3: Cash Position Lookup — Retrieved current Cash Balance of RM 40,000 from the Cash Flow Runway page.

Step 4: Ratio Analysis — Calculated that an RM 2,106.00 outflow represents 5% of your current liquidity (RM 40,000).

Step 5: Decision Logic — At 5% exposure, the recommendation is 'Pay Now' (Risk: LOW). After payment, RM 37,894 would remain.

Confidence & Impact

Confidence: 90%

Quantified Impact

RM 2,106.00 outflow repres…

Full Reasoning

Unstructured data (invoice): Vendor = Global Tech Solutions Inc., Amount = RM 2,106.00, Due = None → Structured data (bank): Cash balance = RM 40,000 → Ratio analysis: Invoice is 5% of available cash → Decision: Pay Now — Invoice amount (RM 2,106.00) is 5% of cash balance (RM 40,000). Remaining after payment: RM 37,894. Sufficient liquidity to pay immediately.

SME Finance AI

Navigate

- Dashboard Overview
- Cash Flow Runway
- AI Insight Generator
- Document Scanner
- Decision Explainability

Decision Explainability

Suggestion: Pay Now — Invoice of RM 2,106.00

Why did the AI suggest this?

TECHNOLOGY STACK

CORE LANGUAGE & FRONTEND

PYTHON

PRIMARY BACKEND LANGUAGE FOR LOGIC, DATA PROCESSING, AND API INTEGRATION.

STREAMLIT

FRAMEWORK UTILIZED FOR THE DEVELOPMENT OF THE INTERACTIVE, HIGH-PERFORMANCE WEB DASHBOARD.

TECHNOLOGY STACK

ARTIFICIAL INTELLIGENCE (AI) & LARGE LANGUAGE MODELS (LLM)

Primary Engine

GEMINI 2.5 FLASH

UTILIZED FOR INITIAL HIGH-CONTEXT DOCUMENT PARSING AND COMPLEX REASONING.

Secondary Failover

Z.AI GLM-4-FLASH

ACTS AS THE SECONDARY INTELLIGENCE LAYER TO ENSURE CONTINUITY IF THE PRIMARY API REACHES RATE LIMITS OR BECOMES UNAVAILABLE.

Local Fallback Logic

A RULE-BASED LOCAL EXTRACTION ENGINE THAT PROCESSES DATA USING REGEX AND PYTHON LOGIC IF BOTH CLOUD LLMS FAIL, ENSURING THE USER IS NEVER LEFT WITHOUT A RESULT.

TECHNOLOGY STACK

DATA PROCESSING & EXTRACTION

PYMUPDF

ADVANCED LIBRARY FOR HIGH-SPEED PDF PARSING AND UNSTRUCTURED TEXT EXTRACTION FROM FINANCIAL DOCUMENTS.

JSON SCHEMA

EMPLOYED FOR THE "DECISION EXPLAINABILITY PANEL" TO ENSURE DATA TRANSPARENCY AND REASONING TRACEABILITY.

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THE IMPACT

THE SYSTEM IMPROVES FINANCIAL PLANNING BY PROVIDING ACCURATE INSIGHTS, SCENARIO SIMULATIONS, AND DATA-DRIVEN DECISION SUPPORT, ENABLING SME OWNERS TO PLAN THEIR FINANCES MORE EFFECTIVELY.

THE SYSTEM HELPS PREVENT CASH SHORTAGES BY ALLOWING USERS TO FORECAST CASH FLOW, EVALUATE POTENTIAL RISKS, AND TAKE PROACTIVE ACTIONS BEFORE FINANCIAL ISSUES ARISE.

THE SYSTEM ENHANCES BUSINESS SUSTAINABILITY BY SUPPORTING INFORMED DECISION-MAKING, REDUCING FINANCIAL RISKS, AND PROMOTING LONG-TERM OPERATIONAL STABILITY.

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OUR FUTURE PLANS

Integrating Real-time Bank Data

For automatic financial tracking

Enhancing AI Accuracy

Using historical data learning

Multi-user Collaboration

For business team use

Enabling Long-term Financial Forecasting

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WITH

SMEZY,

WE TURN
UNCERTAINTY INTO CLARITY

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THANK YOU

SO MUCH!